

José Ferreira

 [jpferreira.me](https://github.com/jpferreira)  jose@jpferreira.me  [jpmvferreira](https://orcid.org/jpmvferreira)  InspireHEP



EDUCATION

- PhD in Physics** 08/2023 - PRESENT
University of Aveiro
Supervisors: Miguel Zilhão, Carlos Herdeiro and Raimon Luna
- MSc in Physics** 09/2020 - 09/2022
Faculty of Sciences of the University of Lisbon
Supervisors: Nelson Nunes and Tiago Barreiro
- BSc in Physics** 09/2017 - 07/2020
Faculty of Sciences of the University of Lisbon

RESEARCH INSTITUTES

- Gr@v** 08/2023 - PRESENT
Member of the Gravitation group at the *Center for R&D in Mathematics and Applications* of the University of Aveiro, Portugal.
- Institute of Astrophysics and Space Sciences** 09/2021 - 07/2023
Member of the cosmology group at the *Institute of Astrophysics and Space Sciences*, Lisbon node, Portugal.

TEACHING

- Assistant** 09/2025 - PRESENT
University of Aveiro
Courses: Numerical and Statistical Methods
- Assistant** 02/2022 - 07/2023
Faculty of Sciences of the University of Lisbon
Courses: Experimental Physics I, Elements of Physics, Mathematical Methods for Physicists, General Physics

PUBLICATIONS

- Jordan Nicoules, **José Ferreira**, Carlos A. R. Herdeiro, Eugen Radu, and Miguel Zilhão. “Splitting the Gravitational Atom: Instabilities of Black Holes with Synchronized/Resonant Hair”. [arXiv: 2509.20450 \[gr-qc\]](https://arxiv.org/abs/2509.20450).
- José Ferreira**, Tiago Barreiro, José Mimoso, and Nelson J. Nunes. “Testing Λ -free $f(Q)$ cosmology”. *Phys. Rev. D* 108 (6 Sept. 2023), p. 063521. [10.1103/PhysRevD.108.063521](https://arxiv.org/abs/10.1103/PhysRevD.108.063521).
- José Ferreira**, Tiago Barreiro, José Mimoso, and Nelson J. Nunes. “Forecasting $F(Q)$ cosmology with Λ CDM background using standard sirens”. *Phys. Rev. D* 105 (12 June 2022), p. 123531. [10.1103/PhysRevD.105.123531](https://arxiv.org/abs/10.1103/PhysRevD.105.123531).

COMMUNICATIONS

TALKS

- “Dynamics of Black Holes with Resonant Hair”. 18th-19th of December, 2025. XVIII Black Holes Workshop, Lisbon, Portugal (Links: [pptx](#))
- “Black Holes as Particle Accelerators”. 24th of October, 2025. Annual Meeting of PhD Students from Maths Department, University of Aveiro, Portugal (Links: [pptx](#))
- “Splitting the (Gravitational) Atom”. 21th-25th of July, 2025. New Frontiers in NR 2025, Mallorca, Spain. (Links: [pptx](#), [pdf](#))
- “(In)stability of Q-Hairy Black Holes”. 20th-22th of May, 2025. NewFunFICO Meeting, Valencia, Spain. (Links: [pptx](#))
- “Non-linear Superradiance of Charged Black Holes: Beyond Spherical Symmetry”. 18th-22th of November, 2024. VI Amazonian Symposium on Physics, Belém, Brazil. (Links: [pptx](#), [pdf](#))

“Using Standard Sirens to test non-metric theories of gravity”. 19th-20th of December, 2023. XVI Black Holes Workshop, Porto, Portugal. (Links: [pptx](#), [pdf](#))

“Forecasting $F(Q)$ Cosmology with Λ CDM Background using Standard Sirens”. 4th-6th of May, 2022. Iberian Cosmology Meeting 2022, Barcelona, Spain. (Links: [recording](#), [pdf](#), [pptx](#))

SEMINARS

“Fate of Black Holes with Resonant Hair”. 12th of December, 2025. Faculty of Sciences of the University of Lisbon, Portugal (Links: [pptx](#))

“Cosmological Tests on Non-Metricity Based Theories of Gravity”. 20th of September, 2023. Gr@v, Aveiro, Portugal. (Links: [recording](#), [pptx](#), [pdf](#))

POSTERS

“Splitting the (Gravitational) Atom”. 21th-25th of July, 2025. New Frontiers in NR 2025, Mallorca, Spain. (Links: [pdf](#))

“Testing Λ -Free $f(Q)$ Cosmology”. 3th-5th of May, 2023. Dark Matter and Stars, Lisbon, Portugal. (Links: [pdf](#))

>_ DIGITAL SKILLS

High Level Languages Julia, Python, Mathematica

Multimedia Editing GIMP, Photoshop, Inkscape, Kdenlive, Audacity

Low Level Languages C, Fortran, CUDA

Web Development Quarto, Hugo

I am an avid Linux user, with experience using it on personal machines, servers, clusters and microcontrollers. I work mostly on a terminal based ecosystem, creating and making use of terminal utilities to develop my projects.

👤 ACADEMIC ACTIVITIES

Supervisor at the IAstro Summer Internships 2023

07/2023

Institute of Astrophysics and Space Sciences

Supervisor of two undergraduate projects that consisted on constraining modified gravity cosmological models with current and forecast datasets coming from different sources.

Podcast “Physically Speaking”

12/2020 - 12/2022

📺 [Fisicamente Falando](#) 🎧 [Fisicamente Falando](#) 📘 [fisicamentefalando.nfef](#) 📷 [fisicamente_falando_](#)

A student driven podcast created during the pandemic where we interviewed faculty members to speak about their lives, both inside and outside of academia. I am in charge of the audio and video editing and also take part in some of the interviews.

Member of the Student Body for Physicists and Physical Engineers

10/2020 - 10/2021

🌐 [nfef-fcul.com](#) 📘 [NFEFFCUL](#) 📷 [nfeffcul](#)

I was in charge for the development of the digital tools of the student body of physicists and physical engineers. This included the website, the online repository (which is made up of student provided content) and the Discord channel.

📡 OUTREACH

European Researchers’ Night 2025

26/09/2025

Coordinator of the participation of Gr@v in the European Researcher’s Night in Aveiro, Portugal. Using digital content, we explained the physics behind Gravitational Waves and lensing around compact objects.

“Echoes and Shadows: How to Detect Black Holes”

03/05/2025

🔗 [Presentation \(.pptx\)](#)

A Presentation, for high school students, on what are black holes and how to find them.

Project “Physics on the loose”

01/2023 - 07/2023

A project, hosted at the Faculty of Sciences of the University of Lisbon, with a room full of interactive experiments to illustrate several fundamental concepts in Physics.

“When a Black Hole Falls Inside... Another Black Hole”

15/04/2023

🔗 [Presentation \(.pptx\)](#)

Presentation of an Ignite style talk in an event organized by the Institute of Astrophysics and Space Sciences that brings space sciences to cities far away from the main urban areas.